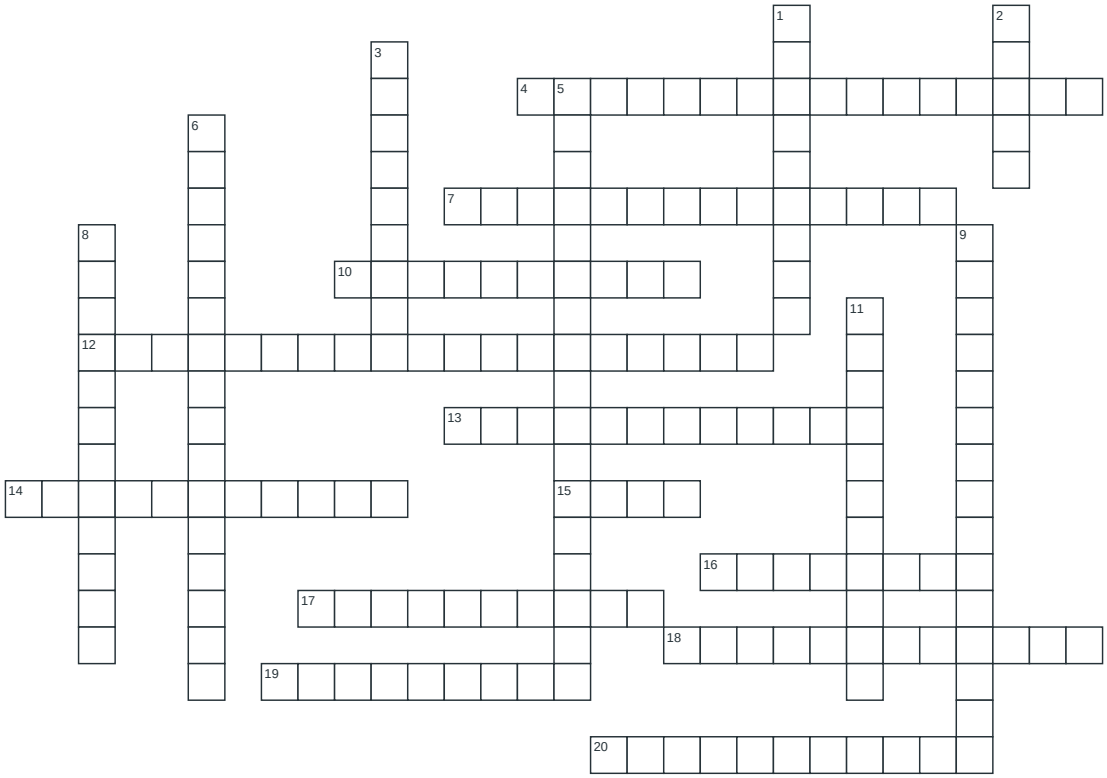
Engineering Design Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

4. Limit such as cost, time, or materials that restricts a design

7. Improving one feature while accepting a disadvantage in another

10. Testable proposed explanation or prediction.

12. Factor deliberately changed by the investigator.

13. Describes data expressed as numbers or measurements.

14. Describes observations about qualities rather than numerical amounts.

15. Recorded observations and measurements from an investigation.

16. Condition kept the same across comparison groups.

17. Number of individuals or observations included in a study.

18. Comparison group that does not receive the tested treatment.

ACROSS (continued)

19. Repeating a test-and-revision cycle

20. Repeating an investigation to check whether results are consistent.

DOWN

1. Ordered steps used to carry out an investigation.

2. One repetition of an experimental procedure.

3. Preliminary version built to test a design

5. Design develops solutions to needs; science investigates explanations of phenomena

6. Difference between a measured value and the quantity’s true value.

8. Improving performance within stated criteria and constraints

9. Feature or level of performance a solution should achieve

11. Information gathered with senses or scientific instruments.

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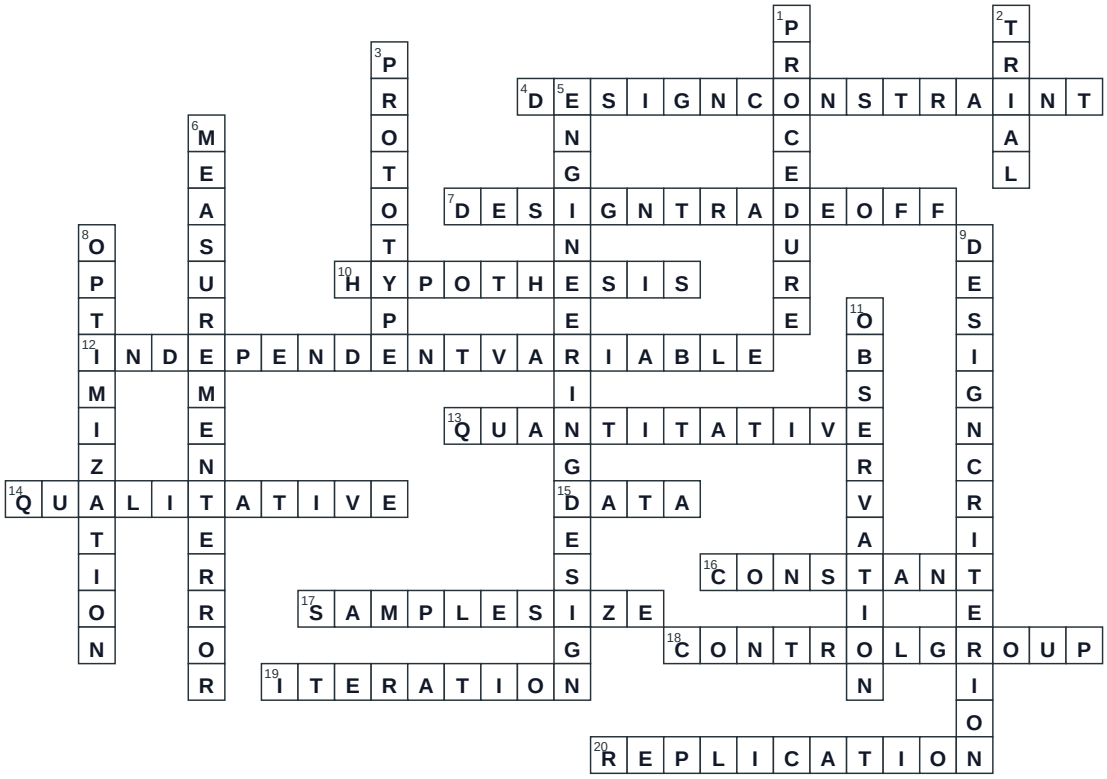
Hard | 20 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

4. Design Constraint
7. Design Tradeoff
10. Hypothesis
12. Independent Variable
13. Quantitative
14. Qualitative
15. Data
16. Constant
17. Sample size
18. Control Group
19. Iteration

ACROSS (continued)

20. Replication

DOWN

1. Procedure
2. Trial
3. Prototype
5. Engineering Design
6. Measurement Error
8. Optimization
9. Design Criterion
11. Observation